

PACKAGE INSERT – CLINT® 100mg Tablets

1. NAME OF THE MEDICINAL PRODUCT

CLINT 100 mg tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 100 mg Allopurinol.

Excipient(s) with known effect: lactose monohydrate.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Tablet for oral administration. White, flat, round, scored tablet with diameter 9.5mm

4. CLINICAL PARTICULARS

4.1. Therapeutic indications

For the reduction of urate levels in the body whether of idiopathic nature or in association with other conditions (neoplastic conditions Lesch-Nyhan syndrome, renal failure, renal calculus formation, diuretic therapy, psoriasis).

4.2. Posology and method of administration

Adults: The initial dose is 100-200mg once daily. The maintenance dose is 200-600mg/daily. Maximum single dose is 300 mg. It has rarely been found necessary to exceed 900mg/day. Should the daily dosage exceed 300mg and gastrointestinal intolerance be manifested, a divided doses regimen may be appropriate. The dose should be adjusted by monitoring serum uric acid and/or urinary uric acid levels at appropriate intervals until the desired effect is attained, which may take one to three weeks.

Children under 15 years: 10 to 20 mg/kg body weight/day up to a maximum of 400mg daily. Use in children is rarely indicated, except in malignant conditions (especially leukemia) and certain enzyme disorders such as Lesch-Nyhan syndrome.

Initiation of therapy: in the early stages of treatment with Allopurinol, as with the uricosuric agents, an acute attack of gouty arthritis may be precipitated. Therefore it is advisable to give a prophylactic dose of a suitable anti-inflammatory agent or colchicines (0.5mg three times a day) for at least one month.

Use with uricocurics: As Allopurinol does not interfere with the action of uricosuric agents, they may be given concurrently. When changing from uricosuric therapy to Allopurinol, 1-3 weeks overlap of treatments is recommended to ensure a continuous hypouricaemic effect.

Use in neoplasia: When applying CLINT to prevent uric acid nephropathy in neoplastic conditions, it is advisable to start treatment with Allopurinol before cytotoxic therapy.

Dose recommendations in impaired renal functions: Since Allopurinol and its metabolites are excreted via the kidney, impairment of renal function may lead to retention of the drug and its metabolites with consequent prolongation of action. Thus, the amount and frequency of the dosage may require reduction as indicated by monitoring serum uric acid levels.

The following schedule is provided for guidance in adults:

If creatinine clearance exceeds 20ml/minute, give standard dose.

If creatinine clearance is between 10 and 20ml/minute, give 100-200mg/day.

If creatinine clearance is between 2 and 10ml/minute, give 100mg/day or at longer intervals.

Dose recommendations in renal dialysis: Allopurinol and its metabolites are removed by renal dialysis. If frequent dialysis is required, an alternative schedule of 300-400mg Allopurinol after each dialysis with none in the interim should be considered.

4.3. Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

4.4. Special warnings and precautions for use

- Allopurinol should not be used for the treatment of an acute attack of gout; additionally, allopurinol therapy should not be initiated for any purpose during an acute attack. However, allopurinol is continued when acute attacks occur in patients already receiving the drug and the acute attack is treated separately.
- Treatment should be stopped immediately if any skin reactions or other signs of hypersensitivity develop. A cautious reintroduction at a low dose may be attempted when mild skin reactions have cleared;
- Hypersensitivity syndrome, SJS and TEN
Allopurinol hypersensitivity reactions can manifest in many different ways, including maculopapular exanthema, hypersensitivity syndrome (also known as DRESS) and SJS/TEN. These reactions are clinical diagnoses, and their clinical presentations remain the basis for decision making. If such reactions occur at any time during treatment, allopurinol should be withdrawn immediately. Rechallenge should not be undertaken in patients with hypersensitivity syndrome and SJS/TEN. Corticosteroids may be beneficial in overcoming hypersensitivity skin reactions.
- HLA-B*5801 allele
The HLA-B*5801 allele has been shown to be associated with the risk of developing allopurinol related hypersensitivity syndrome and SJS/TEN. The frequency of the HLA-B*5801 allele varies widely between ethnic populations: up to 20% in Han Chinese population, about 12% in the Korean population and 1-2% in individuals of Japanese or European origin. The use of genotyping as a screening tool to make decisions about treatment with allopurinol has not been established. If the patient is a known carrier of HLA-B*5801, the use of allopurinol may be considered if the benefits are thought to exceed risks. Extra vigilance for signs of hypersensitivity syndrome or SJS/TEN is required and the patient should be informed of the need to stop treatment immediately at the first appearance of symptoms.
- Dosage should be reduced in renal impairment and in hepatic impairment; extra monitoring may be necessary. Care is advised in patients being treated for hypertension or cardiac insufficiency who may have concomitant renal impairment. To reduce the risk of renal xanthine deposition, an adequate fluid intake (2 to 3 litres daily) is required. In addition, neutral or slightly alkaline urine may be desirable. SJS/TEN can still occur in patients who are found to be negative for HLA-B*5801 irrespective of their ethnic origin.
- Allopurinol should be used with caution in nursing mothers as it has been reported to be distributed into breast milk.
- Chronic renal impairment
Patients with chronic renal impairment may be at increased risk of developing hypersensitivity reactions including SJS/TEN associated with allopurinol. Extra vigilance for the signs of hypersensitivity syndrome or SJS/TEN is required and the patient should be informed of the need to stop treatment immediately and permanently at the first appearance of symptoms
- Hepatic or renal impairment
Reduced doses should be used in patients with hepatic or renal impairment. Patients under treatment for hypertension or cardiac insufficiency, for example with diuretics or ACE inhibitors, may have some concomitant impairment of renal function and allopurinol should be used with care in this group. Asymptomatic hyperuricaemia per se is generally not considered an indication for use of Allopurinol. Fluid and dietary modification with management of the underlying cause may correct the condition. Acute gouty attacks: Allopurinol treatment should not be started until an acute attack of gout has completely subsided, as further attacks may be precipitated. In the early stages of treatment with Allopurinol, as with uricosuric agents, an acute attack of gouty arthritis may be precipitated. Therefore it is advisable to give prophylaxis with a suitable anti-inflammatory agent or colchicine for at least one month. The literature should be consulted for details of appropriate dosage and precautions and warnings. If acute attacks develop in patients receiving allopurinol, treatment should continue at the same dosage while the acute attack is treated with a suitable anti-inflammatory agent.
- Xanthine deposition
In conditions where the rate of urate formation is greatly increased (e.g. malignant disease and its treatment, Lesch-Nyhan syndrome) the absolute concentration of xanthine in urine could, in rare cases, rise sufficiently to allow deposition in the urinary tract. This risk may be minimised by adequate hydration to achieve optimal urine dilution.
- Impaction of uric acid renal stones
Adequate therapy with Allopurinol will lead to dissolution of large uric acid renal pelvic stones, with the remote possibility of impaction in the ureter.

Allopurinol should be discontinued at the first appearance of skin rash or other signs which may indicate an allergic reaction. Hypersensitivity to allopurinol usually appears after some weeks of therapy, and more rarely immediately after beginning treatment. In some instances, a skin rash may be followed by more severe reactions such as exfoliative, urticarial and purpuric lesion as well as Stevens-Johnson syndrome, and/or generalized vasculitis, irreversible hepatotoxicity and even death.

CLINT contains lactose monohydrate. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

4.5. Interactions with other medicinal products and other forms of interaction

Cytostatics

With administration of allopurinol and cytostatics (e.g. cyclophosphamide, doxorubicin, bleomycin, procarbazine, alkyl halogenides), blood dyscrasias occur more frequently than when these active substances are administered alone.

Blood count monitoring should therefore be performed at regular intervals.

Aluminium hydroxide

If aluminium hydroxide is taken concomitantly, allopurinol may have an attenuated effect. There should be an interval of at least 3 hours between taking both medicines.

Coumarin anticoagulants: There have been rare reports of increased effects of warfarin and other coumarin anticoagulants when co-administered with allopurinol, therefore, all patients receiving anticoagulants must be carefully monitored.

Azathioprine and 6-mercaptopurine: Azathioprine is metabolised to 6-mercaptopurine which is inactivated by the action of xanthine oxidase. When 6-mercaptopurine or azathioprine is given concurrently with Allopurinol, only one-quarter of the usual dose of 6-mercaptopurine or azathioprine should be given because inhibition of xanthine oxidase will prolong their activity.

Vidarabine (Adenine arabinoside): Evidence suggests that the plasma half life of vidarabine is increased in the presence of allopurinol. When the two products are used concomitantly extra vigilance is necessary, to recognise enhanced toxic effects.

Salicylates and uricosuric agents: Oxipurinol, the major active metabolite of allopurinol, is excreted by the kidney in a similar way to urate. Hence drugs with uricosuric activity such as probenecid or large doses of salicylates may accelerate the excretion of oxipurinol. This may decrease the therapeutic activity of allopurinol (but the significance needs to be assessed in each case.).

Chlorpropamide: If allopurinol is given concomitantly with chlorpropamide when renal function is poor, there may be an increased risk of prolonged hypoglycaemic activity, because allopurinol and chlorpropamide may compete for excretion in the renal tubule.

Phenytoin: Allopurinol may inhibit hepatic oxidation of phenytoin but the clinical significance has not been demonstrated.

Theophylline: Inhibition of the metabolism of theophylline has been reported. The mechanism of the interaction may be explained by xanthine oxidase being involved in the biotransformation of theophylline in man. Theophylline levels should be monitored in patients starting or increasing allopurinol therapy.

Ampicillin / amoxicillin: An increase in the frequency of skin rash has been reported among patients receiving ampicillin or amoxicillin concurrently with allopurinol compared with patients who are not receiving both drugs. The cause of the reported association has not been established. However, it is recommended that in patients receiving allopurinol, an alternative to ampicillin or amoxicillin is used where available.

Ciclosporin: Reports suggest that the plasma concentration of ciclosporin may be increased during concomitant treatment with allopurinol. The possibility of enhanced ciclosporin toxicity should be considered if the drugs are

co-administered.

Cyclophosphamide, doxorubicin, bleomycin, procarbazine, mechloroethamine: Enhanced bone marrow suppression by cyclophosphamide and other cytotoxic agents has been reported among patients with neoplastic disease (other than leukaemia), in the presence of allopurinol. However, in a well-controlled study of patients treated with cyclophosphamide, doxorubicin, bleomycin, procarbazine and/or mechloroethamine (chlormethine hydrochloride) allopurinol did not appear to increase the toxic reaction of these cytotoxic agents.

Didanosine: In healthy volunteers and HIV patients receiving didanosine, plasma didanosine C_{max} and AUC values were approximately doubled with concomitant allopurinol treatment (300 mg daily) without affecting terminal half life. Co-administration of these 2 drugs is generally not recommended. If concomitant use is unavoidable, a dose reduction of didanosine may be required, and patients should be closely monitored.

Diuretics

An interaction between allopurinol and furosemide that results in increased serum urate and plasma oxypurinol concentrations has been reported.

An increased risk of hypersensitivity has been reported when allopurinol is given with diuretics, in particular thiazides, especially in renal impairment.

Angiotensin-converting-enzyme (ACE) inhibitors

An increased risk of hypersensitivity has been reported when allopurinol is given with ACE inhibitors especially in renal impairment.

4.6. Fertility, pregnancy and lactation

Pregnancy

There is inadequate evidence of safety of Allopurinol in human pregnancy, although it has been in wide use for many years without apparent ill consequence. Use in pregnancy only when there is no safer alternative and when the disease itself carries risk for the mother or unborn child.

Breast-feeding

Allopurinol and its metabolite oxipurinol is excreted in human breast milk. Allopurinol during breast-feeding is not recommended. Concentrations of 1.4mg/litre allopurinol and 53.7 mg/litre oxipurinol have been demonstrated in breast milk from woman taking Allopurinol 300 mg/day. However, there are no data concerning the effects of allopurinol or its metabolites on the breast-fed baby.

4.7. Effects on ability to drive and use machines

No reports.

4.8. Undesirable effects

For this product, there is no modern clinical documentation which can be used as support for determining the frequency of undesirable effects. Undesirable effects may vary in their incidence depending on the dose received and also when given in combination with other therapeutic agents.

The frequency categories assigned to the adverse drug reactions below are estimates: for most reactions, suitable data for calculating incidence are not available. Adverse drug reactions identified through post-marketing surveillance were considered to be rare or very rare. Adverse reactions in association with Allopurinol are rare in the overall treated population and mostly of a minor nature. The incidence is higher in the presence of renal and/or hepatic disorder.

Table 1 Undesirable effects		
System Organ Class	Frequency	Adverse reaction
Infections and infestations	Very rare	Furuncle
Blood and lymphatic system disorders	Very rare	Agranulocytosis ¹ Aplastic anaemia ¹ Thrombocytopenia ¹
Immune system disorders	Uncommon	Hypersensitivity ²

	Very rare	Angioimmunoblastic T-cell lymphoma ³ , Anaphylactic reaction
Metabolism and nutrition disorders	Very rare	Diabetes mellitus Hyperlipidaemia
Psychiatric disorders	Very rare	Depression
Nervous system disorders	Very rare	Coma Paralysis Ataxia Neuropathy peripheral Paraesthesia Somnolence Headache Dysgeusia
Eye disorders	Very rare	Cataract Visual impairment Maculopathy
Ear and labyrinth disorders	Very rare	Vertigo
Cardiac disorders	Very rare	Angina pectoris Bradycardia
Vascular disorders	Very rare	Hypertension
Gastrointestinal disorders	Uncommon	Vomiting ⁴ Nausea ⁴
	Very rare	Haematemesis Steatorrhoea Stomatitis Change of bowel habit
Hepatobiliary disorders	Uncommon	Liver function test abnormal ⁵
	Rare	Hepatitis (including hepatic necrosis and granulomatous hepatitis) ⁵
Skin and subcutaneous tissue disorders	Common	Rash
	Rare	Stevens-Johnson syndrome / toxic epidermal necrolysis ⁶
	Very rare	Angioedema ⁷ Drug eruption Alopecia Hair colour changes
Renal and urinary disorders	Very rare	Haematuria Azotaemia
Reproductive system and breast disorders	Very rare	Infertility male Erectile dysfunction Gynaecomastia
General disorders and administration site conditions	Very rare	Oedema Malaise Asthenia Pyrexia ⁸
Investigations	common	blood thyroid stimulating hormone increased ⁹

1 Very rare reports have been received of thrombocytopenia, agranulocytosis and aplastic anaemia, particularly in individuals with impaired renal and/or hepatic function, reinforcing the need for particular care in this group of patients.

2 A delayed multi-organ hypersensitivity disorder (known as hypersensitivity syndrome or DRESS) with fever, rashes, vasculitis, lymphadenopathy, pseudo lymphoma, arthralgia, leucopenia, eosinophilia, hepato-splenomegaly, abnormal liver function tests, and vanishing bile duct syndrome (destruction and disappearance of the intrahepatic bile ducts) occurring in various

combinations. Other organs may also be affected (e.g. liver, lungs, kidneys, pancreas, myocardium, and colon). If such reactions do occur, it may be at any time during treatment, allopurinol should be withdrawn IMMEDIATELY AND PERMANENTLY.

Rechallenge should not be undertaken in patients with hypersensitivity syndrome and SJS/TEN. Corticosteroids may be beneficial in overcoming hypersensitivity skin reactions. When generalised hypersensitivity reactions have occurred, renal and/or hepatic disorder has usually been present particularly when the outcome has been fatal.

3 Angioimmunoblastic T-cell lymphoma has been described very rarely following biopsy of a generalised lymphadenopathy. It appears to be reversible on withdrawal of Allopurinol tablets.

4 In early clinical studies, nausea and vomiting were reported. Further reports suggest that this reaction is not a significant problem and can be avoided by taking Allopurinol tablets after meals.

5 Hepatic dysfunction has been reported without overt evidence of more generalised hypersensitivity.

6 Skin reactions are the most common reactions and may occur at any time during treatment. They may be pruritic, maculopapular, sometimes scaly, sometimes purpuric and rarely exfoliative, such as Stevens-Johnson syndrome and toxic epidermal necrolysis (SJS/TEN). The highest risk for SJS and TEN, or other serious hypersensitivity reactions, is within the first weeks of treatment. The best results in managing such reactions come from early diagnosis and immediate discontinuation of any suspect drug. Allopurinol tablets should be withdrawn immediately should such reactions occur. After recovery from mild reactions, Allopurinol tablets may, if desired, be re-introduced at a small dose (e.g. 50 mg/day) and gradually increased. If the rash recurs, Allopurinol tablets should be permanently withdrawn as more severe hypersensitivity may occur (see Immune system disorders). If SJS/TEN, or other serious hypersensitivity reactions cannot be ruled out, DO NOT re-introduce allopurinol due to the potential for a severe or even fatal reaction. The clinical diagnosis of SJS/TEN remains the basis for decision making. If such reactions occur at any time during treatment, allopurinol should be withdrawn immediately and permanently.

7 Angioedema has been reported to occur with and without signs and symptoms of a more generalised hypersensitivity reaction.

8 Fever has been reported to occur with and without signs and symptoms of a more generalised Allopurinol tablets hypersensitivity reaction (see Immune system disorders).

9. The occurrence of increased thyroid stimulating hormone (TSH) in the relevant studies did not report any impact on free T4 levels or had TSH levels indicative of subclinical hypothyroidism.

4.9. Overdose

Ingestion of up to 22.5 g allopurinol without adverse effect has been reported. Symptoms and signs including nausea, vomiting, diarrhoea and dizziness have been reported in a patient who ingested 20g allopurinol. Recovery followed general supportive measures. Massive absorption of Allopurinol may lead to considerable inhibition of xanthine oxidase activity, which should have no untoward effects unless affecting concomitant medication, especially with 6-mercaptopurine and/or azathioprine. Adequate hydration to maintain optimum diuresis facilitates excretion of allopurinol and its metabolites. If considered necessary haemodialysis may be used.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: Preparations inhibiting uric acid production, ATC code: M04AA01

Allopurinol is a xanthine-oxidase inhibitor. Allopurinol and its main metabolite oxipurinol lower the level of uric acid in plasma and urine by inhibition of xanthine oxidase, the enzyme catalyzing the oxidation of hypoxanthine to xanthine and xanthine to uric acid. In addition to the inhibition of purine catabolism in some but not all hyperuricaemic patients, de novo purine biosynthesis is depressed via feedback inhibition of hypoxanthine-guanine phosphoribosyltransferase. Other metabolites of allopurinol include allopurinol-riboside and oxipurinol-7 riboside.

5.2. Pharmacokinetic properties

Absorption:

Allopurinol is active when given orally and is rapidly absorbed from the upper gastrointestinal tract. Studies have detected allopurinol in the blood 30-60 minutes after dosing. Estimates of bioavailability vary from 67% to 90%. Peak plasma levels of allopurinol generally occur approximately 1.5 hours after oral administration of Allopurinol, but fall rapidly and are barely detectable after 6 hours. Peak levels of oxipurinol generally occur after 3-5 hours after oral administration of Allopurinol and are much more sustained.

Distribution:

Allopurinol is negligibly bound by plasma proteins and therefore variations in protein binding are not thought to

significantly alter clearance. The apparent volume of distribution of allopurinol is approximately 1.6 litre/kg which suggests relatively extensive uptake by tissues. Tissue concentrations of allopurinol have not been reported in humans, but it is likely that allopurinol and oxipurinol will be present in the highest concentrations in the liver and intestinal mucosa where xanthine oxidase activity is high.

Biotransformation

The main metabolite of Allopurinol tablets is oxipurinol. Other metabolites of allopurinol include allopurinol-ribose and oxipurinol-7-ribose.

Elimination:

Approximately 20% of the ingested allopurinol is excreted in the faeces. Elimination of allopurinol is mainly by metabolic conversion to oxipurinol by xanthine oxidase and aldehyde oxidase, with less than 10% of the unchanged drug excreted in the urine. Allopurinol has a plasma half-life of about 1 to 2 hours.

Oxipurinol is a less potent inhibitor of xanthine oxidase than allopurinol, but the plasma half-life of oxipurinol is far more prolonged. Estimates range from 13 to 30 hours in man. Therefore effective inhibition of xanthine oxidase is maintained over a 24 hour period with a single daily dose of Allopurinol. Patients with normal renal function will gradually accumulate oxipurinol until a steady-state plasma oxipurinol concentration is reached. Such patients, taking 300 mg of allopurinol per day will generally have plasma oxipurinol concentrations of 5-10 mg/litre.

Oxipurinol is eliminated unchanged in the urine but has a long elimination half-life because it undergoes tubular reabsorption. Reported values for the elimination half-life range from 13.6 hours to 29 hours. The large discrepancies in these values may be accounted for by variations in study design and/or creatinine clearance in the patients.

Pharmacokinetics in patients with renal impairment.

Allopurinol and oxipurinol clearance is greatly reduced in patients with poor renal function resulting in higher plasma levels in chronic therapy. Patients with renal impairment, where creatinine clearance values were between 10 and 20ml/min, showed plasma oxipurinol concentrations of approximately 30mg/litre after prolonged treatment with 300 mg allopurinol per day. This is approximately the concentration which would be achieved by doses of 600 mg/day in those with normal renal function. A reduction in the dose of Allopurinol is therefore required in patients with renal impairment.

Pharmacokinetics in elderly patients.

The kinetics of the drug are not likely to be altered other than due to deterioration in renal function (see Pharmacokinetics in patients with renal impairment).

5.3. Preclinical safety data

Not applicable

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

- croscarmellose sodium
- povidone
- lactose monohydrate
- microcrystalline cellulose
- dicalcium phosphate dihydrate
- colloidal silicon dioxide
- magnesium stearate

6.2. Incompatibilities

None reported.

6.3. Shelf life

60 months

6.4. Special precautions for storage

Stored below 30°C in the original packaging

6.5. Nature and contents of container

CLINT® 100mg Tablets are packed in PVC-Alu blisters and available in 1000's (100 x 10's) pack sizes

6.6. Special precautions for disposal

No special requirements for disposal.

7. MARKETING AUTHORISATION HOLDER

Komedic Sdn. Bhd., No. 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor, Malaysia

8. MANUFACTURER

Medochemie Ltd., 1-10 Constantinoupolis Street, 3011 Limassol, Cyprus

9. MARKETING AUTHORISATION NUMBER

MAL19870287AZ

10. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

23/02/1987

11. DATE OF REVISION OF THE TEXT

16 March 2023